

1.To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs

2.To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

Code:

```
priyanka@DESKTOP-HOISO7/  ×  +  v
GNU nano 8.7.1 escape_exec.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <ctype.h>

#define MAX_ARGS 50
#define MAX_LEN 200

int tokenize(char *input, char *args[]) {
    int count = 0;
    int i = 0;

    while (input[i]) {
        while (isspace(input[i]))
            i++;

        if (!input[i])
            break;

        args[count] = malloc(MAX_LEN);
        if (!args[count])
            return -1;

        int j = 0;

        while (input[i] && !isspace(input[i])) {
            if (input[i] == '\\' && input[i + 1]) {
                i++;
                args[count][j++] = input[i++];
            } else {
                args[count][j++] = input[i++];
            }
        }

        if (j >= MAX_LEN - 1)
```

```
        break;
    }

    args[count][j] = '\\0';
    count++;

    if (count >= MAX_ARGS - 1)
        break;
}

args[count] = NULL;
return count;
}

void free_args(char *args[], int count) {
    for (int i = 0; i < count; i++)
        free(args[i]);
}

int main() {
    char input[1000];
    char *args[MAX_ARGS];

    printf("shell> ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\\n")] = '\\0';

    if (strlen(input) == 0) {
        printf("Empty command\\n");
        return 0;
    }

    int count = tokenize(input, args);

    if (count <= 0) {
```

```
priyanka@DESKTOP-HOISO7/  ×  +  ∨
GNU nano 8.7.1                                                     escape_exec.c *
if (count <= 0) {
    printf("Parsing error\n");
    return 1;
}

printf("\nParsed Arguments:\n");

for (int i = 0; i < count; i++)
    printf("%d: %s\n", i + 1, args[i]);

printf("\nLaunching command...\n");

pid_t pid = fork();

if (pid < 0) {
    perror("fork");
    free_args(args, count);
    return 1;
}

if (pid == 0) {
    execvp(args[0], args);

    perror("execvp");
    exit(EXIT_FAILURE);
}

int status;

waitpid(pid, &status, 0);

if (WIFEXITED(status))
    printf("\nChild exited with status %d\n",
        WEXITSTATUS(status));

free_args(args, count);

return 0;
}
```

Output:

```
priyanka@DESKTOP-HOISO7A: ~$ cd ~/process_lab
nano escape_exec.c
priyanka@DESKTOP-HOISO7A:~/process_lab$ gcc -Wall -Wextra -std=c11 escape_exec.c -o escape_exec
priyanka@DESKTOP-HOISO7A:~/process_lab$ ./escape_exec
shell> ls -l

Parsed Arguments:
1: ls
2: -l

Launching command...
total 136
-rwxr-xr-x 1 priyanka priyanka 16592 Aug 18 05:16 escape_exec
-rw-r--r-- 1 priyanka priyanka 2030 Aug 18 05:16 escape_exec.c
-rwxr-xr-x 1 priyanka priyanka 16560 Aug 18 03:54 interactive_shell
-rw-r--r-- 1 priyanka priyanka 3778 Aug 18 03:54 interactive_shell.c
-rwxr-xr-x 1 priyanka priyanka 16368 Aug 18 04:20 parser
-rw-r--r-- 1 priyanka priyanka 2451 Aug 18 04:20 parser.c
-rwxr-xr-x 1 priyanka priyanka 16464 Aug 18 05:05 quotes
-rw-r--r-- 1 priyanka priyanka 3443 Aug 18 05:04 quotes.c
-rwxr-xr-x 1 priyanka priyanka 21536 Aug 18 04:11 task2_1
-rw-r--r-- 1 priyanka priyanka 17340 Aug 18 04:11 task2_1.c

Child exited with status 0
priyanka@DESKTOP-HOISO7A:~/process_lab$ echo hello world
hello world
priyanka@DESKTOP-HOISO7A:~/process_lab$ echo hello\ world
hello world
priyanka@DESKTOP-HOISO7A:~/process_lab$ echo hello\!
hello!
priyanka@DESKTOP-HOISO7A:~/process_lab$ echo hello\@world
hello@world
priyanka@DESKTOP-HOISO7A:~/process_lab$ accxyz
accxyz: command not found
```